



# Snow Mountain Math Adventure

A 5-Day Learn-&-Play Handbook · Year 3 AMC Prep

**For:** Chloe & a friend — learning together on the ski trip

**Each day:** Read & Learn → Warm-up game → Practice → Daily Quiz

**Big Test 1:** after Day 3 · **Final Big Test:** after Day 5

**Time:** 20–25 minutes a day · **Difficulty:** Easy → Hard



## How to Use This Handbook

-  **Read & Learn first:** Each day starts with a short lesson the kids can read out loud together.
-  **Then play:** A warm-up game turns the idea into something you can see and touch.
-  **Practice:** Work through the examples together, talking about each one.
-  **Daily Quiz:** 5 quick questions to check what stuck. +1 snowflake ❄️ for each correct answer — no penalty for wrong answers (just like the real AMC!).
-  **Big Tests:** After Day 3 and after Day 5. These mix everything learned so far. Take your time!
-  **Golden rule:** Whoever answers explains "how I thought about it." Explaining it clearly earns a bonus ❄️.

## Warm-up



## Read & Learn: Adding, Taking Away, and Times

**Adding** means putting groups together. **Subtracting** (taking away) means removing some. When you add, the number gets *bigger*; when you take away, it gets *smaller*.

**Tip:** To add  $8 + 5$ , start at the bigger number (8) and count up 5: 9, 10, 11, 12, 13.

**Times (multiplication)** is a fast way to add the same number many times. "4 groups of 3" is  $3 + 3 + 3 + 3 = 12$ , which we write as  $4 \times 3 = 12$ .

Example: A ski lift has 3 cars. Each car holds 4 people. That's 3 groups of 4 =  $4 + 4 + 4 = 12$  people. Faster:  $3 \times 4 = 12$ .

The "counting-on" trick and the "groups" idea are the two most useful tools in the whole competition.

**Words to know:** **sum** = the answer to an addition · **difference** = the answer to a subtraction · **double** = add a number to itself (double 6 = 12).

**Warm-up game "Count the Cable Cars":** On the lift, count towers and pine trees you pass. Then double it — how many if there were twice as many?

## Practice together

1. There are 8 skiers on the slope. 5 more arrive. How many now?
2. You have 15 snowballs and throw 6. How many are left?
3. Each cable car seats 4 people. How many people in 3 cars? (Use  $\times$ )
4. What is double 7?
5. You start skiing at 9:00 and ski for 2 hours. What time is it now?

**Practice answers:** 1) 13 2) 9 3) 12 ( $3 \times 4$ ) 4) 14 5) 11:00

## Day 1 Daily Quiz (5 questions)

1.  $7 + 6 = ?$
2.  $20 - 8 = ?$
3. 5 groups of 2 is the same as  $5 \times 2 = ?$
4. Which is bigger: double 4, or  $3 + 4$ ?
5. A gondola makes 2 trips before lunch and 3 after. How many trips in total?

**Quiz answers:** 1) 13 2) 12 3) 10 4) double 4 = 8, and  $3+4 = 7$ , so **double 4** is bigger 5) 5

## 17 July Day 2 — The Shape of a Snowflake

### Geometry



#### Read & Learn: Shapes, Sides, and Symmetry

Every flat shape is made of **sides** (straight edges) and **corners** (where two sides meet, also called *vertices*). A triangle has 3 sides and 3 corners. A square has 4 of each. A circle has no straight sides at all!

To find how many sides several shapes have, count one shape's sides, then use times. 3 triangles =  $3 \times 3 = 9$  sides.

**Symmetry** means one half is a mirror of the other. If you fold the shape along a line and both halves match exactly, it has a **line of symmetry**.

Example: A real snowflake has 6 matching points arranged like a star — it is very symmetrical. A square can be folded 4 different ways and still match. The letter **F** can't be folded to match, so it is *not* symmetrical.

**Perimeter** is the distance all the way around the outside of a shape. For a rectangle, add up all four sides.

Rectangle perimeter shortcut:  $(\text{long side} + \text{short side}) \times 2$ .

**Words to know:** **side/edge** = a straight line of a shape · **vertex** = a corner · **symmetry** = both halves match · **perimeter** = distance around the outside.

**Warm-up game "Find Symmetry":** Look at gloves, goggles and cabin windows. Fold a paper napkin — which folds make both halves match? Count how many symmetrical things you can spot in one minute.

#### Practice together

1. A snowflake has 6 points. How many points do 3 snowflakes have? (Use  $\times$ )
2. A triangle has 3 sides and a square has 4 sides. How many sides altogether?
3. Which of these has a line of symmetry? circle · square · letter F
4. How many corners (vertices) does a rectangle have?
5. A cabin window is a rectangle, 5 squares long and 3 squares wide. What is the perimeter (in squares)?

**Practice answers:** 1) 18 ( $3 \times 6$ ) 2) 7 3) circle & square 4) 4 5) 16 ( $((5+3) \times 2)$ )

#### Day 2 Daily Quiz (5 questions)

1. How many sides does a triangle have?
2. 2 squares have how many sides in total?
3. Does a circle have a line of symmetry — yes or no?
4. A square has sides of 4 cm each. What is its perimeter?
5. A shape has 5 sides. It is called a pentagon. How many corners does it have?

**Quiz answers:** 1) 3   2) 8   3) Yes (many, actually!)   4) 16 cm (4×4)   5) 5 (a shape has the same number of sides and corners)

## 17 Day 3 — Shopping at the Ski Resort

### Money



### Read & Learn: Money and Word Problems

Money uses **dollars (\$)** and **cents (¢)**. 100 cents make 1 dollar. When we write **\$6.50**, the part after the dot is the cents — so that's 6 dollars and 50 cents.

**Change is what you get back. Change = money you gave – price. If you pay \$10 for a \$6.50 item, change =  $10 - 6.50 = \$3.50$ .**

**Word problems** hide the maths inside a story. Read it like a detective:

#### 3-step detective plan:

1. **Underline** the numbers and important words (*each, total, left, more, change*).
2. **Decide** the action — add, take away, or times?
3. **Write** the number sentence and solve it.

Watch the key words: "**altogether / total**" usually means add. "**left / change / how many more**" usually means take away. "**each ... how many in all**" usually means times.

**Words to know:** **total** = everything added up · **change** = money given back · **cost** = price you pay · **each** = for every one.

**Warm-up game "Shopkeeper":** One is the shopkeeper, one is the customer. Use real coins. Buy things, work out the total, and hand back the correct change. Then swap roles.

### Practice together

1. A hot chocolate is \$4. Both of you buy one. How much altogether?
2. Chloe has \$10 and buys a \$6.50 sandwich. How much change does she get?
3. A sled rents for \$5 per hour. How much for 3 hours? (Use  $\times$ )
4. A snack pack costs \$3. How many packs can you buy with \$12?
5. Gloves \$8, hat \$7, scarf \$5. With \$20, which TWO items can you buy? List all the possibilities.

**Practice answers:** 1) \$8 2) \$3.50 3) \$15 4) 4 packs ( $12 \div 3$ ) 5) gloves+hat (\$15), gloves+scarf (\$13), hat+scarf (\$12) — all three pairs work!

### Day 3 Daily Quiz (5 questions)

1. How many cents are in one dollar?
2. You buy two \$3 juices. What is the total?
3. You pay \$5 for a \$2 apple. What is your change?
4. Muffins are \$2 each. How much do 4 muffins cost?
5. In the sentence "How many are *left*?", should you add or take away?

**Quiz answers:** 1) 100 2) \$6 3) \$3 4) \$8 ( $4 \times 2$ ) 5) take away (subtract)



## **BIG TEST 1 — After Day 3**

**Mixes Days 1–3.** 10 questions. Take your time, work together or race — your choice. +1 ❄️ each, no penalty for wrong answers.

1.  $9 + 8 = ?$
2. What is  $6 \times 3$ ?
3. How many sides do 4 triangles have altogether?
4. A square's side is 5 cm. What is its perimeter?
5. You have 18 marbles and give away 7. How many are left?
6. Two hot dogs cost \$3 each. What is the total?
7. You pay \$10 for a \$4.50 book. What change do you get?
8. Does the letter "A" have a line of symmetry — yes or no?
9. Double 9 = ?
10. A gondola holds 6 people. How many people fit in 3 gondolas?

**Big Test 1 answers:** 1) 17 2) 18 3) 12 ( $4 \times 3$ ) 4) 20 cm 5) 11 6) \$6 7) \$5.50 8) Yes 9) 18 10) 18 ( $3 \times 6$ )

**Score:** 8–10 = Snow Star ★ · 5–7 = Great work, review the misses · under 5 = re-read those days, no worries!

## Logic



### Read & Learn: Patterns and Clever Thinking

A **pattern** is something that repeats or grows by a rule. To crack a pattern, ask: "*What changes from one to the next?*"

Colours that repeat: red, blue, red, blue... the rule is "repeat every 2". So the next one is red's turn or blue's turn — just keep the cycle going.

Numbers that grow: 5, 10, 15, 20... the rule is "+5 each time". The next number is 25.

**For growing number patterns, find the "jump" (how much it goes up each time), then keep jumping.**

**Logic puzzles** give you clues, and you use them like stepping stones. Line the facts up in order.

"Anna is faster than Chloe. Chloe is faster than Lily." Put them in order fastest → slowest: Anna, Chloe, Lily. So Lily is slowest.

**Drawing a little line or list often makes logic puzzles easy to see.**

**Words to know:** **pattern** = a repeating or growing rule · **sequence** = numbers in order following a rule · **predict** = work out what comes next.

**Warm-up game "Who's Fastest":** Make up three skier friends and give two clues about who is faster. Can the other person work out the order? Then swap.

### Practice together

- Slope flags: red, blue, red, blue, red, \_\_\_\_ — what colour comes next?
- Number pattern: 5, 10, 15, 20, \_\_\_\_ — what comes next?
- Pattern: 2, 4, 6, 8, \_\_\_\_ — what is the jump, and what comes next?
- Anna is faster than Chloe. Chloe is faster than Lily. Who is slowest?
- Day 1: 2 runs, Day 2: 4 runs, Day 3: 6 runs. At this rate, how many on Day 5?

**Practice answers:** 1) blue 2) 25 3) jump is +2, next is 10 4) Lily 5) 10 (2,4,6,8,10)

### Day 4 Daily Quiz (5 questions)

- Pattern: 3, 6, 9, 12, \_\_\_\_ — what comes next?
- Pattern: ▲ ● ▲ ● ▲ \_\_\_\_ — what shape is next?
- Pattern: 10, 20, 30, \_\_\_\_ — what comes next?
- Ben has more sweets than Mia. Mia has more than Tom. Who has the most?



5. Pattern: 1, 2, 4, 8, \_\_\_\_ — (the number doubles each time) what comes next?

**Quiz answers:** 1) 15 2) ▲ (triangle) 3) 40 4) Ben 5) 16

## 17 July Day 5 — The Big Mountain Challenge

Mixed



### Read & Learn: Counting Ways & Multi-Step Problems

Some questions ask "**how many different ways?**" The trick is to be organised and list them without missing any or repeating.

Two people each pick a different hat from red, blue, green. Go in order:  
Red-blue, Red-green, Blue-red, Blue-green, Green-red, Green-blue → **6 ways**.

**Being neat and going in order (a to z, small to big) means you never lose a possibility.**

**Multi-step problems** need more than one calculation. Do one step, write down the answer, then use it for the next step.

Lift up = 15 min, ski down = 10 min. One round trip =  $15 + 10 = 25$  min. Three round trips =  $25 \times 3 = 75$  min. (Two steps!)

**Doubling patterns come up a lot: fold a paper once → 2 parts, twice → 4, three times → 8. Each fold doubles.**

**Words to know:** **combination** = one way of choosing · **multi-step** = needs more than one calculation · **estimate** = a smart guess close to the answer.

**Warm-up game "List It All":** With 3 different snacks, how many different pairs can you make? List them neatly and count. (Answer: 3 pairs.)

### Practice together

1. Red, blue and green hats. Each of you wears a different one. How many combinations?
2. The lift up takes 15 min and skiing down takes 10 min. How long is one round trip? And 3 round trips?
3. Each snowflake has 6 points. How many snowflakes to reach 30 points?
4. Fold a paper in half 2 times, then unfold — how many sections? What about 3 folds?
5. You have 3 different snacks. How many different pairs can you choose?

**Practice answers:** 1) 6   2) one trip = 25 min; 3 trips = 75 min   3) 5 ( $30 \div 6$ )   4) 2 folds = 4 sections; 3 folds = 8 sections  
5) 3 pairs

### Day 5 Daily Quiz (5 questions)

1. You have a red and a blue coat, and a black and a white hat. How many coat+hat outfits can you make?
2. A bus ride is 20 min each way. How long is a return trip?
3. Each pizza is cut into 4 slices. How many slices in 3 pizzas?

4. Fold a paper in half once — how many sections when unfolded?
5. Ways to arrange yourself and one friend in a line (who's first)? How many orders?

**Quiz answers:** 1) 4 (red-black, red-white, blue-black, blue-white) 2) 40 min 3) 12 ( $3 \times 4$ ) 4) 2 5) 2 (you first, or friend first)



## FINAL BIG TEST — After Day 5

**Mixes everything, Days 1–5.** 15 questions, easy → hard, just like a real AMC. Give it 30 minutes. +1 ❄️ each, no penalty for wrong answers. Good luck, mountain mathematicians! 🏂

1.  $12 + 7 = ?$
2.  $15 - 9 = ?$
3. What is  $5 \times 4$ ?
4. Double 8 = ?
5. How many sides do 3 squares have altogether?
6. A rectangle is 6 cm long and 2 cm wide. What is its perimeter?
7. Does the letter "H" have a line of symmetry — yes or no?
8. Two cocoas cost \$4 each. You pay with \$10. What change do you get?
9. Pencils are \$2 each. How many can you buy with \$12?
10. Pattern: 4, 8, 12, 16, \_\_\_\_ — what comes next?
11. Pattern: 1, 2, 4, 8, \_\_\_\_ — (doubles each time) what comes next?
12. Sam is taller than Kim. Kim is taller than Jo. Who is shortest?
13. Red, blue, green scarves — you and a friend each take a different one. How many combinations?
14. A chairlift ride up is 12 min and skiing down is 8 min. How long is one round trip?
15. Each star has 5 points. How many stars make 20 points?

**Final Big Test answers:** 1) 19 2) 6 3) 20 4) 16 5) 12 ( $3 \times 4$ ) 6) 16 cm ( $(6+2) \times 2$ ) 7) Yes 8) \$2 9) 6 ( $12 \div 2$ ) 10) 20 11) 16 12) Jo 13) 6 14) 20 min 15) 4 ( $20 \div 5$ )

**Score:** 13–15 = 🏆 Snow Mountain Math Star! · 9–12 = 🥈 Brilliant, so close! · 5–8 = 🥉 Great effort — review and try again · under 5 = every mathematician re-reads; you've got this!

## Score Tracker

| Task                        | Chloe ❄️ | Friend ❄️ |
|-----------------------------|----------|-----------|
| Day 1 Quiz (/5)             |          |           |
| Day 2 Quiz (/5)             |          |           |
| Day 3 Quiz (/5)             |          |           |
| <b>Big Test 1 (/10)</b>     |          |           |
| Day 4 Quiz (/5)             |          |           |
| Day 5 Quiz (/5)             |          |           |
| <b>Final Big Test (/15)</b> |          |           |
| <b>Total</b>                |          |           |

### Tips for the Grown-up

- **Read the lesson together first.** Let the kids read aloud and put it in their own words — that's where real understanding starts.
- **No rush.** Stop when they're tired. Math is the seasoning, not the main course.
- **Explaining beats answering.** "How did you work it out?" is the most powerful question you can ask.
- **Different paces are normal.** Use team mode when the two kids differ in ability; battle mode when they're close.
- **Hide math in real life.** Cable car numbers, temperatures, times, prices — questions are everywhere on the mountain.

Have a wonderful trip — and happy math!  ❄️ · Plan date: 2026-07-02